

Workshop 5: solutions

1. From Equation (33) in the lecture notes, under the null hypothesis of no formulation effect, we have

$$F = \frac{(D_0 - D_1)/3}{D_1/114} \sim F_{3,114},$$

where $D_0 = 201$ is the deviance for the reduced model and $D_1 = 193$ is the deviance for the full model. The F-test is used because the Gamma model has unknown scale parameter. The full model has six parameters: μ , α_1 , α_2 , and α_3 (we assume, without loss of generality, that α_4 representing the placebo effect, is set to zero for identifiability), β_1 , and β_2 (where, again, one of the three coefficients associated with dose rates levels is set to zero for the same reason). Since $n = 120$ patients, $n - p_1 = 120 - 6 = 114$. Also, $p_1 - p_0$ is simply the difference in the number of parameters between the two models, which is three, due to the fact that we have three identifiable levels for the drug formulation variable. The p-value can therefore be evaluated as follows:

```
F <- ((201-193)/3)/(193/114)
pf(F, 3, 114, lower.tail = FALSE)

## [1] 0.1992741
```

The p-value is large, so there is no evidence against the null hypothesis. We conclude that drug formulation does not appear to affect the change in blood pressure.

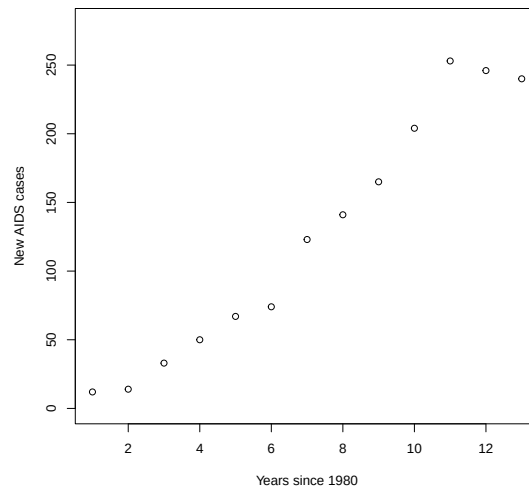
2. Suppose the original GLM has p parameters, and we add one completely irrelevant predictor, so that the enlarged model has $p+1$ parameters. Under the null hypothesis that the smaller model is adequate, the generalized likelihood ratio test result implies, asymptotically

$$D_p^* - D_{p+1}^* \sim \chi_{1(=p+1-p)}^2.$$

Because a χ_1^2 random variable has expected value 1, the expected decrease in scaled deviance is approximately 1.

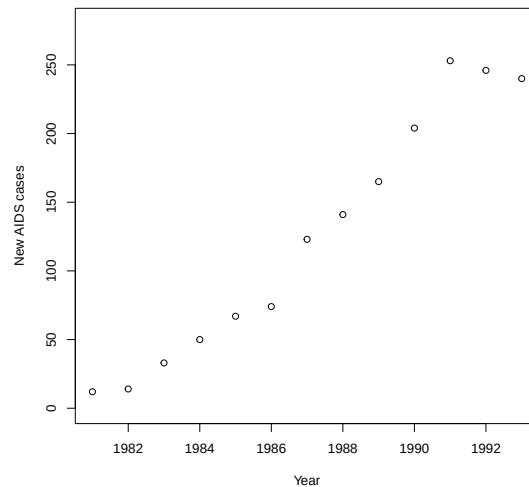
3. (a) One possible way to plot the data is as follows.

```
y <- c(12, 14, 33, 50, 67, 74, 123, 141, 165, 204, 253, 246, 240)
t <- 1:13
plot(t, y, xlab = "Years since 1980", ylab = "New AIDS cases",
     ylim = c(0, 280))
```



We could change the x-axis to be the year instead of the year since 1980.

```
plot(t + 1980, y, xlab = "Year", ylab = "New AIDS cases",
      ylim = c(0, 280))
```



(b) Recall that the model we want to fit is the following:

$$\mathbb{E}(y_i) \equiv \mu_i = \gamma e^{\alpha t_i},$$

which could be turned into a GLM by using the log link so that

$$\log(\mu_i) = \log(\gamma) + \alpha t_i = \beta_1 + \beta_2 t_i, \quad \beta_1 = \log(\gamma), \quad \beta_2 = \alpha.$$

Because the response variable y is a count (the number of new AIDS cases), a Poisson distribution is a sensible choice. Let us fit the model in R.

```
am0 <- glm(y ~ t, family = poisson(link = "log"))
# Alternatively, because the log link is the canonical one
# we could just write
#am0 <- glm(y ~ t, family = poisson)
```

- (c) First, remember that for the Poisson distribution, the scaled deviance coincides with the deviance (because $\phi = 1$). Let us print the summary output of the fitted model to access the deviance.

```
summary(am0)

##
## Call:
## glm(formula = y ~ t, family = poisson(link = "log"))
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  3.140590   0.078247   40.14  <2e-16 ***
## t            0.202121   0.007771   26.01  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 872.206  on 12  degrees of freedom
## Residual deviance:  80.686  on 11  degrees of freedom
## AIC: 166.37
##
## Number of Fisher Scoring iterations: 4
```

The residual deviance is the deviance of the fitted model. Note that we could access it directly by using the command `deviance`:

```
deviance(am0)

## [1] 80.68649
```

We know that under the null hypothesis that the model is ok, the scaled deviance should follow approximately, in the large-sample limit, a chi-squared distribution with 11 degrees of freedom (why 11? there are 13 observations and the model has two parameters). We can then calculate an approximate p-value as follows:

```
pchisq(deviance(am0), df = 11, lower.tail = FALSE)

## [1] 1.086916e-12

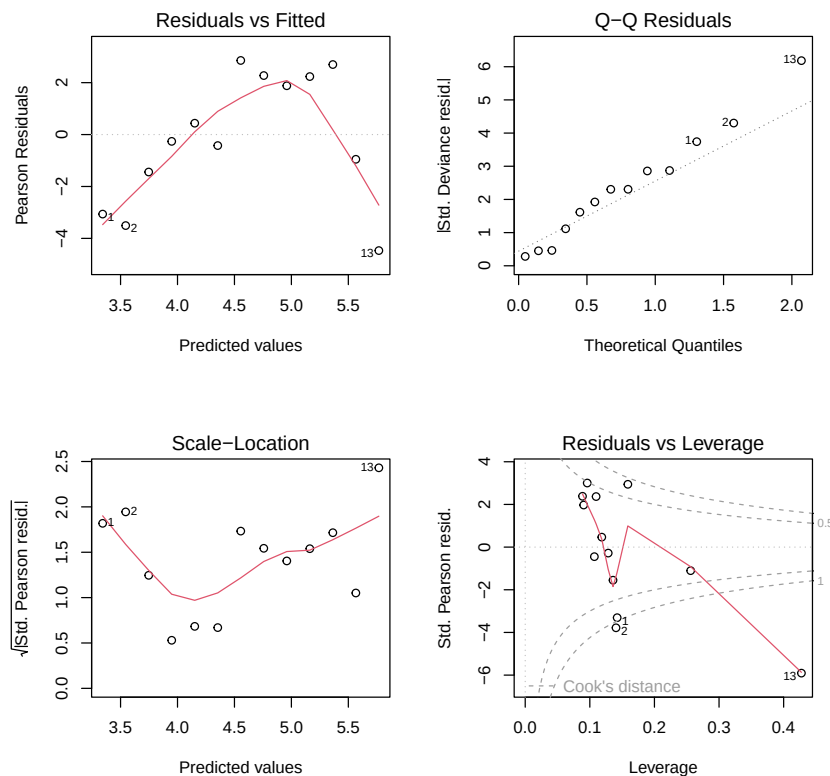
# Equivalent to
#1 - pchisq(deviance(am0), df = 11)
```

The p-value is extremely low and so there is very strong evidence to reject the null hypothesis that the model is compatible with the data. Note that the sample size is low and the result is only approximate, but still, this is a very low p-value. Out of curiosity, just by looking at the deviance,

we should have an idea whether the model is fitting well or not. Recall that if $Z \sim \chi_r^2$, then $E(Z) = r$ and $\text{var}(Z) = 2r$. In this case, we have a deviance of roughly 80, whereas the limiting distribution under the assumption that the model is compatible with the data should have a mean of 11 and a standard deviation of $\sqrt{22} = 4.7$.

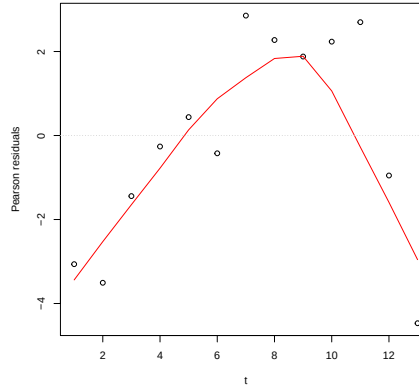
(d) From part (c), we should already anticipate some problematic residual plots.

```
par(mfrow = c(2, 2))
plot(am0)
```



Indeed, the residual plots are worrying. There is a clear pattern in the mean of the residuals, when plotted against the predicted values (and the same would be true if the predicted values were replaced by the fitted values). If we plot the residuals against time we get the same sort of pattern.

```
plot(t, residuals(am0, type = "pearson"), ylab = "Pearson residuals")
lowess_smooth <- lowess(t, residuals(am0, type = "pearson"))
lines(lowess_smooth, col = "red")
abline(h = 0, lty = 3, col = "gray80")
```



It is also worrying the very high influence of the final year's data, evident in the residuals versus leverage plot.

- (e) From the residual plots in (d) it seems sensible to amend the model by adding a quadratic term to obtain:

$$\log(\mu_i) = \beta_1 + \beta_2 t_i + \beta_3 t_i^2.$$

The following code fits this model.

```
aml <- glm(y ~ t + I(t^2), family = poisson(link = "log"))
```

- (f) Let us begin by printing the summary output of this new model, which includes a quadratic effect of time.

```
summary(aml)

##
## Call:
## glm(formula = y ~ t + I(t^2), family = poisson(link = "log"))
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.901459   0.186877  10.175  < 2e-16 ***
## t            0.556003   0.045780  12.145  < 2e-16 ***
## I(t^2)       -0.021346   0.002659  -8.029  9.82e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 872.2058  on 12  degrees of freedom
## Residual deviance:  9.2402  on 10  degrees of freedom
## AIC: 96.924
##
## Number of Fisher Scoring iterations: 4
```

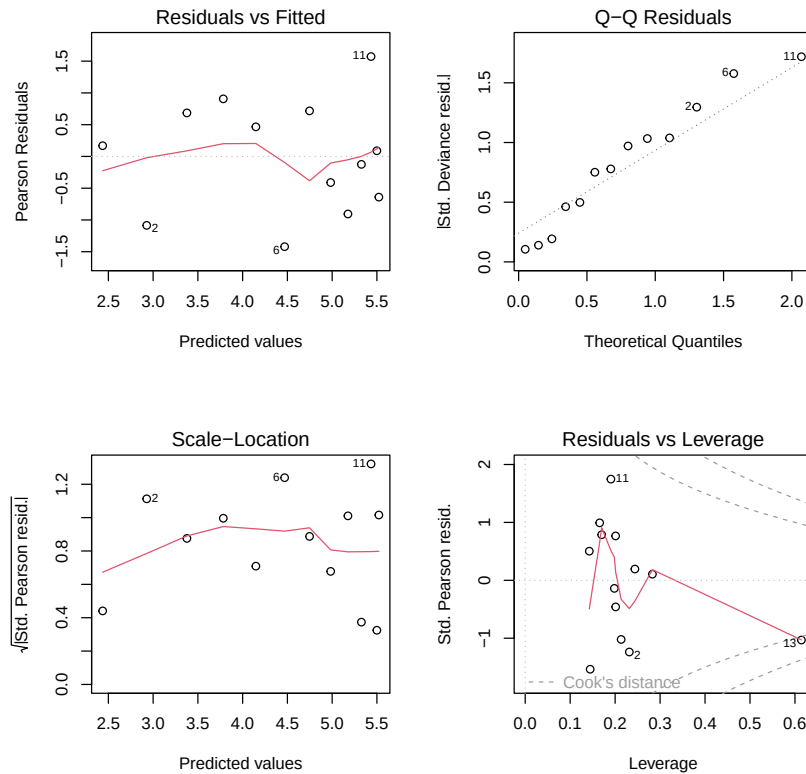
We can notice that the deviance of the fitted model is now much lower and close to what is expected for a χ^2_{10} random variable. Indeed, the approximate p-value does not lead to a rejection of the null hypothesis (that the model is ok).

```
pchisq(deviance(am1), df = 10, lower.tail = FALSE)

## [1] 0.5094652
```

We now inspect the residual plots.

```
par(mfrow = c(2,2))
plot(am1)
```



The residual plots are now much improved: the clear trend in the mean has disappeared, the vertical spread of the residuals is reasonably even, and the influence of point 13 is greatly reduced. Regarding the significance of the terms, the reported p-values are very low, that is, there is clear evidence that each parameter is significantly different from zero.

- (g) We have the AIC included in the summary of each model but we can also extract it easily using the function `AIC`:

```
AIC(am0, am1)

##      df      AIC
## am0  2 166.36982
## am1  3  96.92358
```

AIC is much lower for model `am1` and so based on this criterion this should be the preferred model.

- (h) First, residual plots are much better for model `am1`, i.e., for the model with a quadratic trend. It would not have been reliable to base any conclusions on model `am0` in light of its residual

plots. Second, all coefficients are firmly significant in model `am1`, including β_3 , providing clear evidence that `am1` is preferable to `am0`. The AIC also prefers model `am1` to model `am0`.

As an example, although it was not needed here, in order to check if model `am1` should be preferred to model `am0`, we could have used a generalized likelihood ratio test as follows.

```
anova(am0, am1, test = "Chisq")

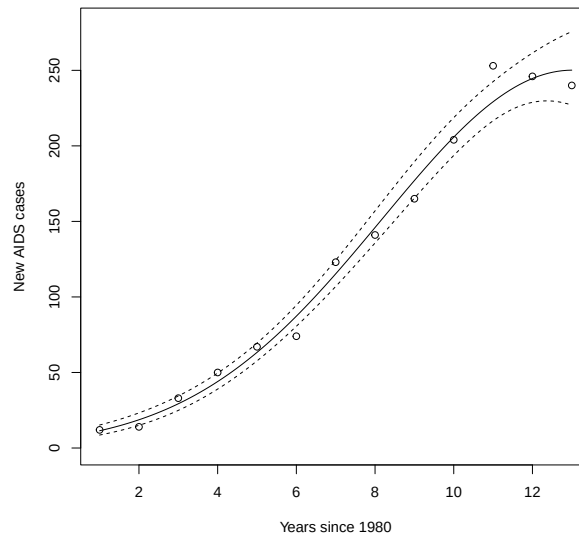
## Analysis of Deviance Table
##
## Model 1: y ~ t
## Model 2: y ~ t + I(t^2)
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1         11      80.686
## 2         10       9.240   1    71.446 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Obviously, the conclusion is the same as before: the tiny p-value indicates that `am0` should be firmly rejected in favour of `am1`. The `test = "Chisq"` argument to `anova` is justified because the scale parameter (ϕ) is known for this model: had it been estimated it would be preferable to set `test` to `"F"`.

- (i) From the summary output the quadratic coefficient (β_3) is negative, so exponential rise is slowing with time. We can also confirm this with a plot of the predicted number of cases (solid line) along with 95% confidence limits (dashed line).

```
new.t <- seq(1, 13, length = 100)
fv <- predict(am1, data.frame(t = new.t), se = TRUE)
qz <- qnorm(0.975)

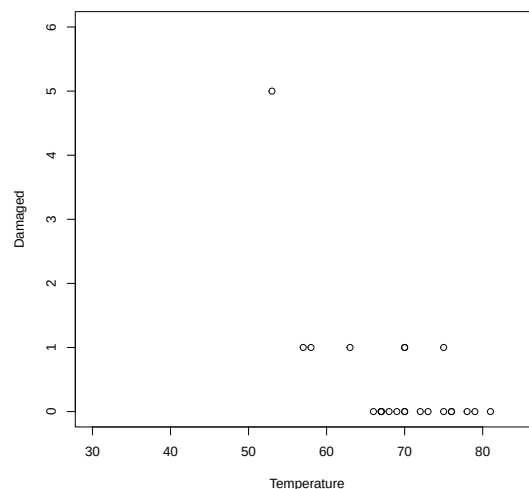
plot(t, y, xlab = "Years since 1980", ylab = "New AIDS cases",
     ylim = c(0, 280))
lines(new.t, exp(fv$fit))
lines(new.t, exp(fv$fit - qz*fv$se.fit), lty = 2)
lines(new.t, exp(fv$fit + qz*fv$se.fit), lty = 2)
```



By the end of the data there is evidence that the count curve is flattening. Notice that by default the `predict` function when passed a `glm` object predicts on the scale of the linear predictor: we have to apply the inverse of the link function to get back onto the original response scale.

4. (a) One possibility to reproduce the plot is as follows.

```
load("orings.rda")
plot(orings$temperature, orings$damaged, ylim = c(0, 6),
     xlim = c(30, 85), xlab = "Temperature", ylab = "Damaged")
```



```
# Another possibility
# with(orings, plot(temperature, damaged,
#                   xlim = c(30, 85), ylim = c(0, 6)))
```

- (b) Let us fit the suggested logistic regression model.


```

oringm <- glm(cbind(damaged, undamaged) ~ temperature,
              binomial(link = "logit"), data = orings)
oringm

##
## Call:  glm(formula = cbind(damaged, undamaged) ~ temperature, family = binomial(
##       data = orings)
##
## Coefficients:
## (Intercept)  temperature
##      11.6630      -0.2162
##
## Degrees of Freedom: 22 Total (i.e. Null);  21 Residual
## Null Deviance:      38.9
## Residual Deviance: 16.91  AIC: 33.67

```

An alternative way of fitting this model is as follows.

```

oringm_alt <- glm(damaged/(damaged + undamaged) ~ temperature,
                  binomial(link = "logit"),
                  weights = damaged + undamaged, data = orings)
oringm_alt

##
## Call:  glm(formula = damaged/(damaged + undamaged) ~ temperature, family = binom
##       data = orings, weights = damaged + undamaged)
##
## Coefficients:
## (Intercept)  temperature
##      11.6630      -0.2162
##
## Degrees of Freedom: 22 Total (i.e. Null);  21 Residual
## Null Deviance:      38.9
## Residual Deviance: 16.91  AIC: 33.67

```

Once we have fitted the model, and even though it was not specifically asked in the question, how should we interpret the parameters in a logistic regression model? In particular, how should we interpret β_2 ? Using the logit link function with the binomial distribution provides a useful interpretation. To understand this, we must first understand the concept of *odds*. If event A has probability p of occurring, then the odds of event A occurring is the ratio of the probability that A occurs to the probability that it does not: $p/(1 - p)$. For example, if the probability of O-ring damage is 0.6, the odds of O-ring damage is $0.6/(1 - 0.6) = 1.5$. Clearly, using the logit link function in a binomial GLM is equivalent to modelling the logarithm of the odds (or the ‘log-odds’). In this specific case, the log odds of O-ring damage as a function of the temperature. We

have the following for a given temperature i , temp_i

$$\begin{aligned}\log\left(\frac{p_i}{1-p_i}\right) &= \log\left(\frac{p_{\text{temp}_i}}{1-p_{\text{temp}_i}}\right) = \beta_1 + \beta_2 \text{temp}_i, \\ \log\left(\frac{p_{\text{temp}_i+1}}{1-p_{\text{temp}_i+1}}\right) &= \beta_1 + \beta_2(\text{temp}_i + 1).\end{aligned}$$

Therefore,

$$\log\left(\frac{p_{\text{temp}_i+1}}{1-p_{\text{temp}_i+1}}\right) - \log\left(\frac{p_{\text{temp}_i}}{1-p_{\text{temp}_i}}\right) = \beta_1 + \beta_2(\text{temp}_i + 1) - (\beta_1 + \beta_2 \text{temp}_i) = \beta_2.$$

Thus, β_2 represents the change in log odds of O-ring damage associated with a unit increase in temperature. Further note that,

$$\log\left(\frac{\frac{p_{\text{temp}_i+1}}{1-p_{\text{temp}_i+1}}}{\frac{p_{\text{temp}_i}}{1-p_{\text{temp}_i}}}\right) = \beta_2 \Rightarrow \frac{\frac{p_{\text{temp}_i+1}}{1-p_{\text{temp}_i+1}}}{\frac{p_{\text{temp}_i}}{1-p_{\text{temp}_i}}} = e^{\beta_2} \Rightarrow \frac{p_{\text{temp}_i+1}}{1-p_{\text{temp}_i+1}} = e^{\beta_2} \frac{p_{\text{temp}_i}}{1-p_{\text{temp}_i}}.$$

So the odds of O-ring damage at $\text{temp}_i + 1$ is e^{β_2} times the odds at temp_i . In this particular example $\hat{\beta}_2 = -0.216 \Rightarrow e^{-0.216} = 0.806$ and so the odds of O-ring damage at $\text{temp}_i + 1$ is 0.806 times the odds at temp_i (so a unit temperature increase leads to a decrease in the odds of O-ring damage). We can also easily attach a confidence interval to this statement.

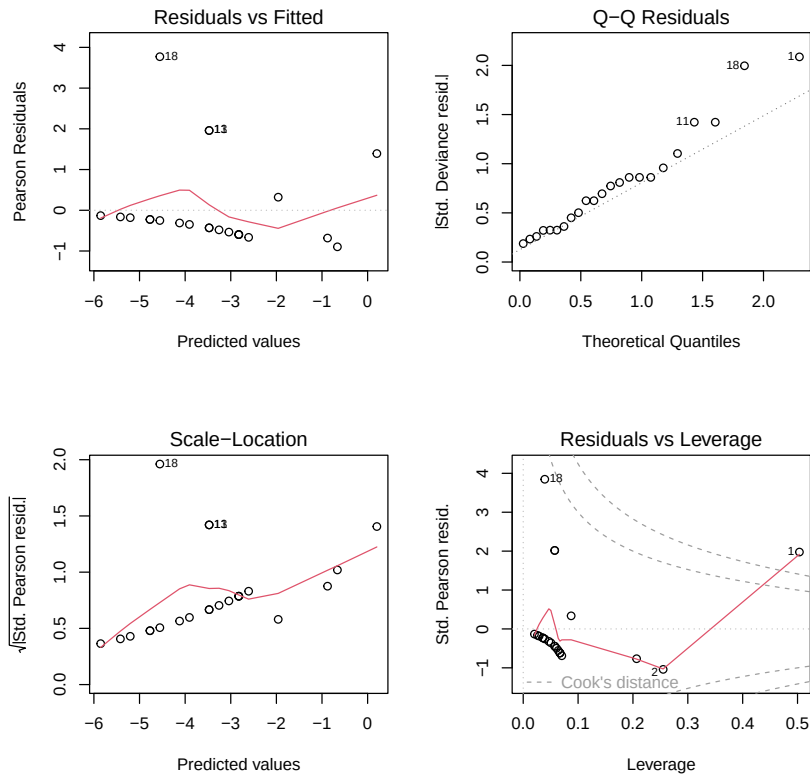
```
exp(confint.default(oringm)[2,])

##      2.5 %      97.5 %
## 0.7258160 0.8940366
```

That is, the odds of O-ring damage at $\text{temp}_i + 1$ is 0.806 (0.726, 0.894) times the odds at temp_i . Note that if $\beta_2 = 0$ then we would have $e^{\beta_2} = 1$ and there would be no effect on the odds of O-ring damage.

- (c) As observed from the plot in part (a), the number of damaged O-rings at each launch temperature is essentially either zero or one. Consequently, when residuals are plotted against the fitted values or the linear predictor, observations with $y_i = 0$ and $y_i = 1$ lie on separate curves, because the residuals are functions of y_i and $\hat{\mu}_i$. Instead of forming a random scatter, the residuals cluster along these curves, producing a patterned, non-continuous display. This makes residual plots less informative for assessing model adequacy in this example. To convince ourselves.

```
par(mfrow = c(2, 2))
plot(oringm)
```



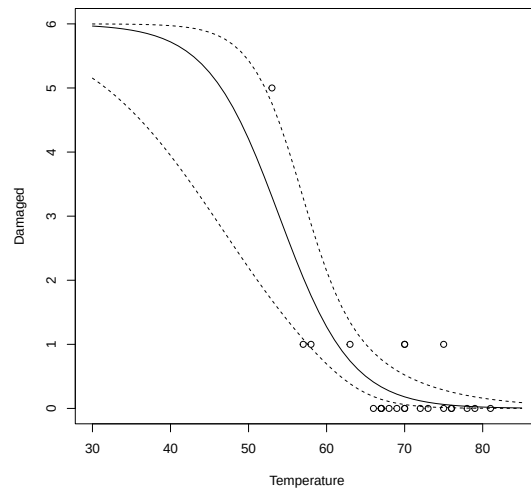
- (d) Let us construct the plot asked in the question. Remember that predictions will be on the scale of the linear predictor, so that we need to convert back to the probability scale by applying the inverse of the logit function. We were told in the exercise sheet how this can be easily implemented in R. Note that this is implementing nothing more than

$$p_i = \frac{e^{\beta_1 + \beta_2 \text{temp}_i}}{1 + e^{\beta_1 + \beta_2 \text{temp}_i}} = \frac{1}{1 + e^{-\beta_1 - \beta_2 \text{temp}_i}}.$$

Further note that $\mathbb{E}(y_i) = 6 \times p_i$.

```
new_temperature <- 30:85
fv <- predict(oringm, data.frame(temperature = new_temperature),
              se = TRUE)
qz <- qnorm(0.975)
ilogit <- binomial()$linkinv

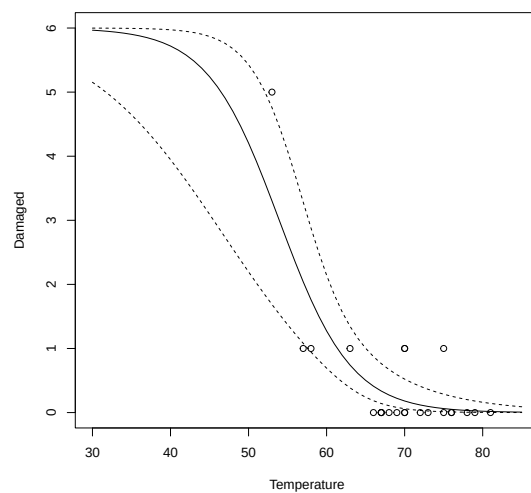
plot(oringm$temperature, oringm$damaged, ylim = c(0, 6),
     xlim = c(30, 85), xlab = "Temperature", ylab = "Damaged")
lines(new_temperature, ilogit(fv$fit)*6, col = 1)
uc <- fv$fit + qz*fv$se.fit
lines(new_temperature, ilogit(uc)*6, col = 1, lty = 2)
lc <- fv$fit - qz*fv$se.fit
lines(new_temperature, ilogit(lc)*6, col = 1, lty = 2)
```



As mentioned before, instead of the `ilogit` function we could have just implemented everything ‘manually’.

```
probs <- exp(fv$fit)/(1 + exp(fv$fit))
probs_l <- exp(fv$fit - qz*fv$se.fit)/(1 + exp(fv$fit - qz*fv$se.fit))
probs_u <- exp(fv$fit + qz*fv$se.fit)/(1 + exp(fv$fit + qz*fv$se.fit))

plot(orings$temperature, orings$damaged, ylim = c(0, 6),
      xlim = c(30, 85), xlab = "Temperature", ylab = "Damaged")
lines(new_temperature, probs*6, col = 1)
lines(new_temperature, probs_l*6, col = 1, lty = 2)
lines(new_temperature, probs_u*6, col = 1, lty = 2)
```



The fit seems reasonable. Further we have that the deviance is 16.912 on 21 degrees of freedom which seems acceptable. Let us confirm by computing the approximate p-value.

```
pchisq(deviance(oringm), df = 21, lower.tail = FALSE)

## [1] 0.7164099
```

Indeed, there is a quite high probability of a χ^2_{21} r.v. being as large as 16.912.

- (e) The expected number of O-ring failures (95% CI) at the Challenger launch temperature of 31 degrees F is 5.958 (5.067, 5.998). This suggests, in fact, a disaster, with six (lower bound of five) of the six O-rings failing. Note, however, that we are extrapolating far beyond the observed data: the minimum launch temperature in the dataset is 53 F and most launches occurred at temperatures above 65 F.

```
6*logit(fv$fit[2])

##          2
## 5.958205

6*logit(fv$fit[2] - qz*fv$se.fit[2])

##          2
## 5.066826

6*logit(fv$fit[2] + qz*fv$se.fit[2])

##          2
## 5.998397
```

- (f) We know that observations with a Cook's distance above one have the potential to impact the results. Let's check the exact value of Cook's distance for the first observation.

```
cooks.distance(oringm)[1]

##          1
## 1.980532
```

A Cook's distance of almost 2! We should remove this observation from the dataset, rerun the analysis, and check if the conclusions are maintained. In particular, does the model still predict a disaster at 31F? I will not repeat the full analysis here, but you are encouraged to try it yourself!

5. (a) Let us start by fitting fit the independence model as required.

```
al <- data.frame(y = c(435, 147, 375, 134),
                 gender = as.factor(c("F", "F", "M", "M")),
                 belief = as.factor(c(1, 0, 1, 0)))

al

##      y gender belief
## 1 435      F      1
## 2 147      F      0
## 3 375      M      1
## 4 134      M      0

alm0 <- glm(y ~ gender + belief, family = poisson, data = al)
model.matrix(alm0)
```

```
##      (Intercept) genderM belief1
## 1           1           0           1
## 2           1           0           0
## 3           1           1           1
## 4           1           1           0
## attr(,"assign")
## [1] 0 1 2
## attr(,"contrasts")
## attr(,"contrasts")$gender
## [1] "contr.treatment"
##
## attr(,"contrasts")$belief
## [1] "contr.treatment"

alm0

##
## Call:  glm(formula = y ~ gender + belief, family = poisson, data = al)
##
## Coefficients:
## (Intercept)      genderM      belief1
##      5.010      -0.134       1.059
##
## Degrees of Freedom: 3 Total (i.e. Null);  1 Residual
## Null Deviance:      272.7
## Residual Deviance: 0.162  AIC: 35.41
```

We know that the model in the worksheet is not identifiable and R, as confirmed by the model matrix, sets females and believers as the baseline categories by default. The deviance of this independence model is 0.162, which should be compared against a chi-squared distribution with one degree of freedom. The p-value (computed below) does not provide evidence to reject the null hypothesis that the model fits the data adequately.

```
pchisq(deviance(alm0), df = 1, lower.tail = FALSE)

## [1] 0.6873263
```

We can also look at the fitted values (which are at the scale of the response) and compare them to the entries of the contingency table.

```
fitted(alm0)

##      1      2      3      4
## 432.099 149.901 377.901 131.099
```

The model fit appears to be quite good.

- (b) In order to check a possible association between gender and after life belief we need to fit a model with an interaction between these two factors and compare against the independence model. The interaction model can be written as

$$\log(\mu_i) = \tau + \beta_k + \delta_j + \zeta_{kj} \text{ if } y_i \text{ is gender } k \text{ and faith } j,$$

where ζ_{kj} is the *interaction parameter* (of course, we know that this model is not identifiable, but we know how to make it identifiable and, in particular, what R will do). Let us now implement this interaction model.

```
alm1 <- glm(y ~ gender*belief, family = poisson, data = al)
alm1

##
## Call:  glm(formula = y ~ gender * belief, family = poisson, data = al)
##
## Coefficients:
##      (Intercept)          genderM          belief1  genderM:belief1
##           4.99043          -0.09259           1.08491          -0.05583
##
## Degrees of Freedom: 3 Total (i.e. Null);  0 Residual
## Null Deviance:      272.7
## Residual Deviance: 3.331e-14  AIC: 37.25
```

To test whether there is evidence for an interaction between gender and believe, we test the null hypothesis that model alm0 is correct against the more general alternative that alm1 is correct, using an analysis of deviance.

```
anova(alm0, alm1, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: y ~ gender + belief
## Model 2: y ~ gender * belief
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         1      0.162
## 2         0      0.000  1    0.162   0.6873
```

A p-value of 0.69 suggests that there is no evidence to reject model alm0 and the hypothesis of no association between gender and belief in the afterlife. Notice that, in fact, the model with the interaction is the saturated model, which is why its deviance is numerically zero. Therefore, there was no real need to fit it and compare it explicitly with the independence model; in this case, we could have simply examined the deviance of the independence model.

6. We begin by fitting the required binomial regression models using the three different link functions.

```
logdose <- c(1.691, 1.724, 1.755, 1.784, 1.811, 1.837, 1.861, 1.884)
dead <- c(6, 13, 18, 28, 52, 53, 61, 60)
n <- c(59, 60, 62, 56, 63, 59, 62, 60)
alive <- n - dead

fit_logit <- glm(cbind(dead, alive) ~ logdose,
                 family = binomial(link = "logit"))
# Alternative way
#fit_logit_alt <- glm(dead/n ~ logdose,
#                      family = binomial(link = "logit"), weights = n)
fit_probit <- glm(cbind(dead, alive) ~ logdose,
```

Observed	Logit	Probit	Comp.log.log
6	3.5	3.4	5.7
13	9.8	10.7	11.3
18	22.4	23.4	20.9
28	33.9	33.8	30.3
52	50.0	49.6	47.7
53	53.3	53.4	54.2
61	59.2	59.7	61.1
60	58.8	59.2	59.9

```
family = binomial(link = "probit"))
fit_cloglog <- glm(cbind(dead, alive) ~ logdose,
  family = binomial(link = "cloglog"))
```

We can start by comparing the expected number of deaths under the three models to the observed number of deaths.

```
df <- data.frame("Observed" = dead,
  "Logit" = n*fit_logit$fitted.values,
  "Probit" = n*fit_probit$fitted.values,
  "Comp log-log" = n*fit_cloglog$fitted.values)
require(kableExtra)
knitr::kable(df, escape = FALSE, digits = 1) %>%
  kable_styling(position = "center")
```

The fitted values obtained from the binomial model with the complementary log-log link function appear to match the observed values more closely, compared to those from the binomial model using either a logit or a probit link function. Note that all three models use the same number of parameters: two, β_0 and β_1 . Since the models with different link functions are not nested, they cannot be compared using likelihood ratio tests. However, we can compare them using the AIC criterion.

```
AIC(fit_logit, fit_probit, fit_cloglog)
```

```
##           df      AIC
## fit_logit    2 41.31361
## fit_probit    2 40.18499
## fit_cloglog    2 33.71237
```

The AIC is lowest for the binomial model with the complementary log-log link function. Let us further inspect this model.

```
summary(fit_cloglog)

##
## Call:
## glm(formula = cbind(dead, alive) ~ logdose, family = binomial(link = "cloglog"))
```



```
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -39.522      3.236  -12.21  <2e-16 ***
## logdose      22.015      1.797   12.25  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 284.2024  on 7  degrees of freedom
## Residual deviance:   3.5143  on 6  degrees of freedom
## AIC: 33.712
##
## Number of Fisher Scoring iterations: 4
```

We have that the deviance is 3.51 on 6 degrees of freedom and therefore

```
pchisq(3.51, df = 6, lower.tail = FALSE)

## [1] 0.7426388
```

shows that the probability of observing a deviance at least as large as 3.51 under the fitted model is quite high, so there is no evidence of lack of fit.

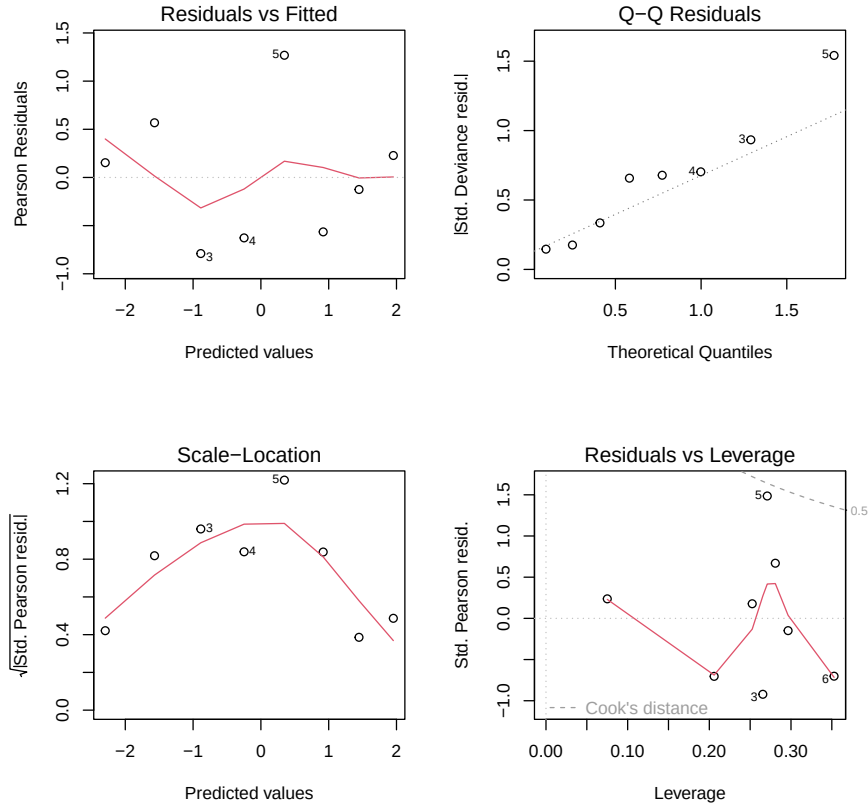
We have that

$$\log(-\log(1 - \hat{p}_i)) = -39.522 + 22.015x_i \Rightarrow \hat{p}_i = 1 - e^{-e^{-39.522+22.015x_i}},$$

which allows us to compute the predicted probability of death at any dose of interest.

Finally, let us look at the residual plots, although we have very few observations.

```
par(mfrow = c(2, 2))
plot(fit_cloglog)
```



The trend in the scale-location plot suggests a possible departure from the assumed binomial mean-variance relationship, but with only 8 observations this is difficult to assess reliably.

7. (a) • Writing $m_i, o_i, p_i, v_i, n_i, a_i, e_i$ and r_i for variables in the order presented in the question (excluding `pollib`), then the initial model fitted is

$$m_i \sim \text{Poisson}(\mu_i),$$

$$\log \mu_i = \alpha + \beta_1 o_i + \beta_2 p_i + \beta_3 v_i + \beta_4 n_i + \beta_5 a_i + \beta_6 e_i + \beta_7 r_i + \gamma_j, \quad (1)$$

if country i is liberalization type j , with $j \in \{0, 1, 2\}$.

- The residual plots do not show gross problems, although Sudan and Liberia appear to be influential observations.
- Backward selection is applied, using generalized likelihood ratio tests to compare models with and without individual terms. When comparing nested models 0 and 1 with deviances D_0 and D_1 and with p_1 and p_0 identifiable parameters, respectively, then under the null hypothesis that the smaller model is correct, and asymptotically,

$$D_0 - D_1 \sim \chi^2_{p_1 - p_0}$$

if model 0 is correct, and this is the basis for p-value calculations.

- Terms drop out sequentially until we are left with the model

$$\log \mu_i = \alpha + \beta_1 o_i + \gamma_j \text{ if country } i \text{ is liberalization type } j$$

Actually, the p-value for testing $H_0 : \gamma_j = 0 \forall j$ is above the p-value for other terms that have previously been dropped, so there is an argument for selecting the model depending only on o_i , but perhaps the analyst is really interested in the γ_j 's, and therefore leaves them in.

- (b) The coefficients suggest that political liberalization reduces the number of coups, but the evidence that these effects are real is only marginal. There is a significant positive relationship between years of military rule and number of coups — this is a rather unsurprising effect: more coups is almost always going to lead to more years of military rule.
- (c) The proportion of deviance explained for the selected model is

```
1 - (45.431/79.124)
## [1] 0.4258253
```

- (d) Only in a rather limited sense. The main predictor is number of years under military rule: almost without exception, a country only gets to be under military rule because of a coup. So we're predicting number of coups using a variable that implicitly contains the information that at least one coup happened. There's no meaningful sense in which you could use the model to predict whether a country not in the list would be likely to suffer a coup.
- (e) Using AIC model selection the model

$$\log\{\mathbb{E}(m_i)\} = \alpha + \beta_1 o_i + \beta_2 p_i + \gamma_j, \text{ if country } i \text{ is liberalization type } j,$$

would be selected. Notice the odd fact that the AIC for *the same model* sometimes changes between one call to `drop1` and the next (even though the scale parameter is known to be 1 here). This happens because there are missing values in the predictor variable data (see final model summary for evidence of this). Any country with missing values is simply omitted when model fitting, but as the number of predictors is reduced fewer and fewer countries need to be omitted. When comparing models by AIC (or hypothesis testing) it is important that they are fitted to the same response data, so `drop1` always fits all model alternatives to the response data usable by the largest model. However different calls to `drop1` will use different numbers of countries as the largest model under consideration changes. This explains the apparent discrepancy.

8. (a) As we have done before, we start by writing $f(y^*; \mu, k) = \exp\{\log f(y^*; \mu, k)\}$:

$$\begin{aligned} f(y^*; \mu, k) &= \exp\{\log f(y^*; \mu, k)\} \\ &= \exp\left\{\log \Gamma(ky^* + k) - \log \Gamma(k) - \log \Gamma(ky^* + 1) + ky^* \log\left(\frac{\mu}{\mu + k}\right) + k \log\left(\frac{k}{\mu + k}\right)\right\} \\ &= \exp\left\{\frac{y^* \log\left(\frac{\mu}{\mu + k}\right) - \left[-\log\left(\frac{k}{\mu + k}\right)\right]}{1/k} + [\log \Gamma(ky^* + k) - \log \Gamma(k) - \log \Gamma(ky^* + 1)]\right\} \end{aligned}$$

This is in the exponential family of distributions with:

- $\theta = \log\left(\frac{\mu}{\mu + k}\right) \Rightarrow \mu = \frac{e^\theta}{1 - e^\theta} k.$

- $b(\theta) = -\log\left(\frac{k}{\mu+k}\right)$. Replacing μ by $\frac{e^\theta}{1-e^\theta}k$, we get that

$$\begin{aligned}\mu + k &= \frac{e^\theta}{1-e^\theta}k + k = k\left(\frac{e^\theta}{1-e^\theta} + 1\right) = k\left(\frac{1}{1-e^\theta}\right), \\ \frac{k}{\mu + k} &= \frac{k}{k\left(\frac{1}{1-e^\theta}\right)} = 1 - e^\theta,\end{aligned}$$

and therefore

$$b(\theta) = -\log\left(\frac{k}{\mu+k}\right) = -\log(1 - e^\theta).$$

- $a(\phi) = \frac{1}{k}$.

(b) By the properties of the exponential family we have that

$$\mathbb{E}(Y^*) = b'(\theta) = \frac{e^\theta}{1-e^\theta} = \frac{\mu}{k}.$$

We further have that

$$\mathbb{E}(Y) = \mathbb{E}(kY^*) = k\mathbb{E}(Y^*) = k\frac{\mu}{k} = \mu.$$

We also know that $\text{var}(Y^*) = b''(\theta)a(\phi)$. Let us workout $b''(\theta)$:

$$b''(\theta) = \frac{d}{d\theta} \left(\frac{e^\theta}{1-e^\theta} \right) = \frac{e^\theta}{(1-e^\theta)^2} = \frac{e^\theta}{1-e^\theta} \times \frac{1}{1-e^\theta} = \frac{\mu}{k} \times \left(1 + \frac{\mu}{k}\right),$$

where we note that

$$1 + \frac{\mu}{k} = 1 + \frac{e^\theta}{1-e^\theta} = \frac{1}{1-e^\theta}.$$

Thus,

$$\text{var}(Y^*) = b''(\theta)a(\phi) = \frac{\mu}{k} \times \left(1 + \frac{\mu}{k}\right) \times \frac{1}{k} = \frac{\mu}{k^2} + \frac{\mu^2}{k^3}.$$

We finally have that

$$\text{var}(Y) = \text{var}(kY^*) = k^2\text{var}(Y^*) = k^2 \left(\frac{\mu}{k^2} + \frac{\mu^2}{k^3} \right) = \mu + \frac{\mu^2}{k}.$$

- (c) For the Poisson distribution, the mean and variance are equal. Therefore, when the variance is substantially larger than the mean, the Poisson model is not appropriate. In such cases, the negative binomial distribution may be more suitable, since its variance exceeds the mean and can accommodate overdispersion. Moreover, as $k \rightarrow \infty$, the term $\mu^2/k \rightarrow 0$, and the negative binomial distribution approaches the Poisson distribution as a limiting case.